

I. POWER AND RESPONSIBILITY: THE ETHICAL HACKER

1.1. The Moral Compass of a Hacker

In the field of Cyber Security, technical capability alone is not sufficient to define professionalism. The true distinction between a cybersecurity expert and a cybercriminal lies not in technical skill, but in intent, authorization, and ethical responsibility.

Cybersecurity professionals possess the same technical knowledge that malicious hackers use to compromise systems. They understand vulnerabilities, exploit techniques, and methods of bypassing defenses. However, ethical professionals apply this knowledge for defensive purposes—to identify weaknesses before malicious actors can exploit them.

At Duy Tan University, students are taught that cybersecurity is fundamentally a discipline of trust and responsibility. Technical power must always be guided by ethical principles and legal compliance.

Ethical hacking is based on three essential principles:

- **Authorization:** Security testing must always be performed with explicit permission.
- **Transparency:** Findings must be documented and reported responsibly.
- **Protection:** The objective is always to strengthen systems, not exploit them.

Without these principles, technical expertise can quickly become dangerous.

1.2. The Hacker Classifications

Cybersecurity professionals are commonly classified into three categories based on their intent and methods.

White Hat Hackers

White Hat hackers are authorized security professionals who use their skills to identify vulnerabilities and help organizations strengthen defenses.

Characteristics:

- Operate legally with written authorization
- Conduct penetration testing ethically
- Provide remediation recommendations

Examples include:

- Security consultants
- Penetration testers
- Bug bounty researchers

Their work contributes directly to safer digital systems.

Black Hat Hackers

Black Hat hackers exploit vulnerabilities for malicious purposes such as:

- Financial theft
- Data breaches
- Corporate espionage

- Service disruption

These individuals operate without authorization and intentionally violate legal and ethical boundaries.

Their actions can result in:

- Massive financial losses
- Reputation damage
- National security threats

Grey Hat Hackers

Grey Hat hackers operate in a legally ambiguous space.

They may:

- Discover vulnerabilities without permission
- Report them to the system owner
- Sometimes request compensation

Although their intentions may not be malicious, unauthorized access remains illegal under most legal frameworks, including Vietnamese cybersecurity law.

1.3. The DTU Professional Commitment

Students in the Cyber Security program at Duy Tan University are expected to adopt a strict professional code of conduct.

Core commitments include:

- Never access data without authorization
- Never disrupt digital services intentionally
- Never exploit vulnerabilities for personal gain
- Always report discovered vulnerabilities responsibly

This ethical foundation ensures that graduates contribute positively to the cybersecurity profession.



II. COMPLIANCE & LEGAL CAREER PATHS

2.1. Beyond the Code: Legal Security

Cyber Security is no longer viewed solely as a technical discipline. It has evolved into a legal and regulatory requirement for organizations worldwide.

Modern businesses must comply with laws governing:

- Personal data protection
- Cyber incident reporting
- Digital privacy
- Critical infrastructure security

Failure to comply can result in legal penalties, financial losses, and reputational damage.

This creates a growing demand for professionals who understand both **technology and law**.

2.2. Emerging Career Roles

Data Privacy Officer (DPO)

Responsible for ensuring that organizations process personal data in compliance with legal regulations.

Responsibilities include:

- Monitoring data collection practices
- Conducting privacy assessments
- Managing compliance audits

IT Auditor

Evaluates an organization's technical controls and security policies to ensure regulatory compliance.

Tasks include:

- Reviewing access controls
- Assessing risk management procedures
- Validating policy implementation

Cyber Law Consultant

Bridges the gap between legal systems and cybersecurity technologies.

Their work may involve:

- Advising corporations on legal compliance
- Supporting digital forensic investigations
- Assisting in cybercrime prosecution

2.3. The Economic Impact of Compliance

Regulatory non-compliance can lead to:

- Heavy financial penalties
- Operational restrictions
- Loss of customer trust
- Suspension of business licenses

As digital regulations become stricter, compliance professionals are among the most valuable members of cybersecurity teams.

III. VIETNAM'S CYBERSECURITY LAW & INTERNATIONAL STANDARDS

3.1. The Law on Cybersecurity (2018)

Vietnam's **Law on Cybersecurity**, which came into effect in 2019, is the primary legal framework governing activities in cyberspace.

It addresses key areas including:

Data Localization

Certain categories of user data must be stored on servers located within Vietnam.

This requirement aims to:

- Enhance national data sovereignty
- Facilitate legal investigation
- Improve government oversight

Prohibited Acts

The law clearly prohibits activities such as:

- Unauthorized system access
- Distribution of malicious software
- Cyber sabotage
- Online disinformation threatening public order

Critical Infrastructure Protection

Special protections apply to systems essential to national operations, including:

- Banking systems
- Telecommunications networks
- Energy infrastructure
- Government platforms

3.2. International Comparison: GDPR and Global Standards

To prepare students for international careers, DTU also introduces global legal frameworks such as:

- **GDPR (European Union)**
- **ISO/IEC 27001**
- **NIST Cybersecurity Framework**

Understanding these standards is essential for professionals working with multinational corporations such as:

- Google
- Amazon
- Microsoft
- FPT Software

3.3. Why Legal Knowledge Matters

A technically sound solution may still fail if it violates legal standards.

Cybersecurity professionals must therefore evaluate:

- Technical effectiveness
- Legal compliance
- Ethical implications

IV. DEEP DIVE: INCIDENT REPORTING & LEGAL PROCEDURES

4.1. What Happens When a Breach Occurs?

When a cybersecurity incident occurs, the response process involves more than technical remediation.

Organizations must:

- Investigate the incident
- Preserve evidence
- Notify authorities
- Document all actions

Improper handling can compromise investigations and create legal liabilities

4.2. The Reporting Chain

Detection

The first step involves identifying indicators of compromise such as:

- Unusual login activity
- Unauthorized file access
- Unexpected system behavior

Preservation

Digital evidence must be preserved using forensic methods to maintain integrity.

Examples include:

- Disk imaging
- Hash verification
- Chain-of-custody documentation

Notification

In Vietnam, incidents may need to be reported to:

- A05 (Ministry of Public Security)
- VNCERT/CC

Timely reporting is essential for legal compliance.

4.3. Penalties for Cybercrime

Under the Vietnamese Penal Code, cyber offenses can result in:

- Significant monetary fines
- Criminal prosecution
- Long-term imprisonment

Examples include:

- Unauthorized access to systems
- Malware distribution
- Data theft
- Service disruption attacks

Understanding these penalties reinforces the seriousness of cybersecurity law.

4.4. Practical Lab Exercise

A common academic exercise involves drafting a formal **Security Incident Report**.

Students must include:

- Incident description
- Evidence summary
- Technical findings
- Legal references
- Recommended response actions

This exercise teaches the intersection of technical accuracy and legal clarity.

V. LEARNING OUTCOMES

Upon completing this module, students will be able to:

- Distinguish ethical from unethical hacking practices
- Understand Vietnam's cybersecurity legal framework
- Apply incident reporting procedures
- Evaluate compliance requirements
- Integrate legal awareness into technical decision-making